Volume Iii Bounds Sequences And Series Reference

Contents

Structure of the Real Line

0.1 Real Line One Dimensional

0.2 Order Distance Length

Definition 0.2.1 (Distance on the Real Line)**.** The *distance* on $\mathbb{R}$ is the map $d : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ given by

$$d(x, y) = |x - y|.$$

Definition 0.2.2 (Length of a Bounded Interval)**.** Let $a, b \in \mathbb{R}$ with $a \leq b$. The *length* of any bounded interval with endpoints a and b is

$$\ell = b - a.$$

Theorem 0.2.3 (The Distance Function Is a Metric)**.** *The map $d(x, y) = |x - y|$ satisfies, for all $x, y, z \in \mathbb{R}$: (i) $d(x, y) \geq 0$ with $d(x, y) = 0$ iff $x = y$; (ii) $d(x, y) = d(y, x)$; and (iii) $d(x, z) \leq d(x, y) + d(y, z)$. Hence $(\mathbb{R}, d)$ is a metric space.* *Go to proof.*

0.3 Absolute Value as Distance

Proposition 0.3.1 (Absolute Value Is Distance to the Origin)**.** *For every $a \in \mathbb{R}$, $|a| = d(a, 0)$.* *Go to proof.*

0.4 Intervals as Subsets

Definition 0.4.1 (Bounded Subset of the Real Line)**.** A set $E \subseteq \mathbb{R}$ is *bounded* exactly when there exists $M \geq 0$ such that $|x| \leq M$ for all $x \in E$; equivalently, E is contained in some closed interval $[-M, M]$.

Set Operations on Intervals

Definition 0.4.2 (Union of Intervals)**.** Let $I, J \subseteq \mathbb{R}$ be intervals. Their *union* is the set

$$I \cup J = \{x \in \mathbb{R} : x \in I \text{ or } x \in J\}.$$

Definition 0.4.3 (Intersection of Intervals)**.** Let $I, J \subseteq \mathbb{R}$ be intervals. Their *intersection* is the set

$$I \cap J = \{x \in \mathbb{R} : x \in I \text{ and } x \in J\}.$$

Definition 0.4.4 (Complement of an Interval)**.** Let $I \subseteq \mathbb{R}$ be an interval. Its *complement* in $\mathbb{R}$ is the set

$$I^c = \mathbb{R} \setminus I = \{x \in \mathbb{R} : x \notin I\}.$$

Definition 0.4.5 (Difference of Intervals)**.** Let $I, J \subseteq \mathbb{R}$ be intervals. The *difference* of I by J is the set

$$I \setminus J = \{x \in \mathbb{R} : x \in I \text{ and } x \notin J\} = I \cap J^c.$$

0.5 Neighborhoods and Balls

Definition 0.5.1 (Open Ball on the Real Line)**.** Let $x \in \mathbb{R}$ and $r > 0$. The *open ball* of radius r about x is

$$B_r(x) = \{\, y \in \mathbb{R} : |y - x| < r \,\} = (x - r,\, x + r).$$

Definition 0.5.2 (Neighborhood of a Point)**.** A set $N \subseteq \mathbb{R}$ is a *neighborhood* of $x \in \mathbb{R}$ exactly when there exists $r > 0$ with $B_r(x) \subseteq N$.

0.6 Open Sets

Definition 0.6.1 (Open Set in $\mathbb{R}$)**.** A set $U \subseteq \mathbb{R}$ is *open* exactly when

$$\forall x \in U \; \exists r > 0 \; (B_r(x) \subseteq U).$$

Theorem 0.6.2 (Open Intervals Are Open)**.** *Every open interval (a, b), every open ray (a, ∞) and $(-\infty, b)$, and $\mathbb{R}$ itself are open sets.* *Go to proof.*

Theorem 0.6.3 (Unions and Finite Intersections of Open Sets)**.** *An arbitrary union of open subsets of $\mathbb{R}$ is open, and a finite intersection of open subsets of $\mathbb{R}$ is open.* *Go to proof.*

0.7 Closed Sets

Definition 0.7.1 (Closed Set in $\mathbb{R}$)**.** A set $F \subseteq \mathbb{R}$ is *closed* exactly when its complement $\mathbb{R} \setminus F$ is open.

Theorem 0.7.2 (Closed Sets Contain Their Limit Points)**.** *A set $F \subseteq \mathbb{R}$ is closed if and only if it contains every one of its limit points.* *Go to proof.*

0.8 Interior Exterior Boundary

Definition 0.8.1 (Interior Point)**.** Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. Then x is an interior point of E exactly when

$$\exists r > 0 \; (B_r(x) \subseteq E).$$

Definition 0.8.2 (Exterior Point)**.** Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. Then x is an exterior point of E exactly when

$$\exists r > 0 \; (B_r(x) \subseteq \mathbb{R} \setminus E).$$

Definition 0.8.3 (Boundary Point)**.** Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. Then x is a boundary point of E exactly when

$$\forall r > 0 \; (B_r(x) \cap E \neq \emptyset \;\wedge\; B_r(x) \cap (\mathbb{R} \setminus E) \neq \emptyset).$$

Definition 0.8.4 (Interior of a Set)**.** The *interior* of $E \subseteq \mathbb{R}$, written $\operatorname{int}(E)$, is the set of all interior points of E.

0.9 Limit and Isolated Points

Definition 0.9.1 (Limit Point)**.** Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. Then x is a *limit point* of E exactly when

$$\forall r > 0 \; \exists y \in E \; \big(0 < |y - x| < r\big).$$

Definition 0.9.2 (Isolated Point)**.** Let $E \subseteq \mathbb{R}$ and $x \in E$. Then x is an *isolated point* of E exactly when

$$\exists r > 0 \; \forall y \in E \; \big(|y - x| < r \Rightarrow y = x\big).$$

0.10 Unions and Intersections

Theorem 0.10.1 (Closure Laws for Closed Sets)**.** *An arbitrary intersection of closed subsets of* $\mathbb{R}$ *is closed, and a finite union of closed subsets of* $\mathbb{R}$ *is closed. Go to proof.*

0.11 Covers and Subcovers

Definition 0.11.1 (Open Cover)**.** Let $E \subseteq \mathbb{R}$. An *open cover* of E is a family $\{U_i\}_{i\in I}$ of open subsets of $\mathbb{R}$ with

$$E \subseteq \bigcup_{i\in I} U_i.$$

Definition 0.11.2 (Subcover and Finite Subcover)**.** Let $\{U_i\}_{i\in I}$ be an open cover of E. A *subcover* is a subfamily $\{U_i\}_{i\in J}$ with $J \subseteq I$ that still covers E. It is a *finite subcover* when J is finite.

0.12 Compactness on R

Definition 0.12.1 (Compact Set in $\mathbb{R}$)**.** A set $K \subseteq \mathbb{R}$ is *compact* exactly when every open cover of K admits a finite subcover:

$$\forall \{U_i\}_{i\in I} \text{ open cover of } K\ \exists \text{finite } J \subseteq I \left(K \subseteq \bigcup_{i\in J} U_i\right).$$

Theorem 0.12.2 (Compact Subsets of $\mathbb{R}$ Are Closed and Bounded)**.** *If* $K \subseteq \mathbb{R}$ *is compact, then* K *is closed and bounded. Go to proof.*

0.13 Heine Borel

Theorem 0.13.1 (Closed Bounded Intervals Are Compact)**.** *Every closed bounded interval* $[a, b] \subseteq \mathbb{R}$ *is compact. Go to proof.*

Theorem 0.13.2 (Heine–Borel Theorem for $\mathbb{R}$)**.** *A set* $K \subseteq \mathbb{R}$ *is compact if and only if* K *is closed and bounded. Go to proof.*

Proofs

Bounds

0.14 Completeness Axiom

Axiom of Completeness

Definition 0.14.1 (Least Upper Bound Property)**.** Let $(S, \leq)$ be an ordered set. The set S has the *Least Upper Bound Property* if and only if every nonempty subset of S that is bounded above in S has a supremum in S.

0.14.1 Nested Intervals

Nested Interval Property

Theorem 0.14.2 (Nested Interval Property)**.** *Let (I_n) be a sequence of nonempty closed intervals in $\mathbb{R}$. If $I_{n+1} \subseteq I_n$ for every $n \in \mathbb{N}$, then*

$$\bigcap_{n=1}^{\infty} I_n \neq \varnothing.$$

Go to proof.

0.14.2 Archimedean Property

Archimedean Property

Theorem 0.14.3 (Archimedean Property)**.** *For every real number x, there exists a natural number n such that $n > x$.*

Go to proof.

Corollary 0.14.4 (Archimedean Reciprocal Corollary)**.** *For every $x > 0$ and every $y \in \mathbb{R}$, there exists $n \in \mathbb{N}$ such that $nx > y$.*

Go to proof.

0.14.3 Density

Density

Definition 0.14.5 (Dense Subset)**.** Let $D \subseteq S \subseteq \mathbb{R}$. The set D is *dense in S* if and only if for every $x \in S$ and every $\varepsilon > 0$ there exists $d \in D$ such that $|x - d| < \varepsilon$.

Definition 0.14.6 (Order Dense Subset)**.** Let X be a linearly ordered set and let $A \subseteq X$. The subset A is *order dense in X* if and only if for every $a, b \in X$ with $a < b$ there exists $c \in A$ such that $a < c < b$.

Theorem 0.14.7 (Density of the Rational Numbers in $\mathbb{R}$)**.** *For every $a, b \in \mathbb{R}$ with $a < b$, there exists $q \in \mathbb{Q}$ such that $a < q < b$.* *Go to proof.*

Theorem 0.14.8 (Density of the Irrational Numbers in $\mathbb{R}$). *For every $a, b \in \mathbb{R}$ with $a < b$, there exists $s \in \mathbb{R} \setminus \mathbb{Q}$ such that $a < s < b$.* *Go to proof.*

Corollary 0.14.9 (Rational and Irrational Points in Every Open Interval). *Every open interval in $\mathbb{R}$ contains both a rational number and an irrational number.* *Go to proof.*

0.14.4 Completeness Summary

0.15 Bounds and Extremal Values

0.15.1 Upper and Lower Bounds

Definition 0.15.1 (Bound). Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and let $b \in S$. A *bound* of A means a one-sided comparison between b and every element of A: either $x \leq b$ for all $x \in A$, or $b \leq x$ for all $x \in A$.

Definition 0.15.2 (Upper Bound). Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and let $u \in S$. The element u is an *upper bound* of A if every element of A is less than or equal to u.

Definition 0.15.3 (Lower Bound). Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and let $\ell \in S$. The element ℓ is a *lower bound* of A if every element of A is greater than or equal to ℓ.

Definition 0.15.4 (Bounded Above). Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. The set A is *bounded above* if it has at least one upper bound in S.

Definition 0.15.5 (Bounded Below). Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. The set A is *bounded below* if it has at least one lower bound in S.

Definition 0.15.6 (Bounded Set). Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. The set A is *bounded* if it is bounded above and bounded below.

0.15.2 Suprema and Infima

Definition 0.15.7 (Supremum). Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and let $s \in S$. The element s is the *supremum* of A if s is an upper bound of A and no smaller element of S is an upper bound of A.

Definition 0.15.8 (Infimum). Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and let $i \in S$. The element i is the *infimum* of A if i is a lower bound of A and no larger element of S is a lower bound of A.

Proposition 0.15.9 (Uniqueness of the Supremum). *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $s, t \in S$ are both suprema of A, then $s = t$.*
Go to proof.

Proposition 0.15.10 (Uniqueness of the Infimum). *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $i, j \in S$ are both infima of A, then $i = j$.*

Proposition 0.15.11 (Upper Bound Property of the Supremum). *Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and suppose that $s = \sup A$ exists in S. Then every element of A lies below s:*

$$(\forall x \in A)(x \leq s).$$

Go to proof.

Proposition 0.15.12 (Lower Bound Property of the Infimum). *Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and suppose that $i = \inf A$ exists in S. Then every element of A lies above i:*

$$(\forall x \in A)(i \leq x).$$

Lemma 0.15.13 (Subset Inclusion Preserves Upper Bounds)**.** *Let $A \subseteq B \subseteq S$ be nonempty subsets of an ordered set $(S, \leq)$. If u is an upper bound of B, then u is an upper bound of A.*
Go to proof.

Lemma 0.15.14 (Subset Inclusion Preserves Lower Bounds)**.** *Let $A \subseteq B \subseteq S$ be nonempty subsets of an ordered set $(S, \leq)$. If ℓ is a lower bound of B, then ℓ is a lower bound of A.*

Theorem 0.15.15 (Supremum Is Monotone Under Inclusion)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. If $A \subseteq B$, then $\sup A \leq \sup B$.*
Go to proof.

Theorem 0.15.16 (Infimum Is Monotone Under Inclusion)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded below. If $A \subseteq B$, then $\inf B \leq \inf A$.*

Proposition 0.15.17 (Upper Bound Comparison with the Supremum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $s = \sup A$. For $u \in \mathbb{R}$, the number u is an upper bound of A if and only if $s \leq u$.*
Go to proof.

Proposition 0.15.18 (Lower Bound Comparison with the Infimum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $i = \inf A$. For $\ell \in \mathbb{R}$, the number ℓ is a lower bound of A if and only if $\ell \leq i$.*

Proposition 0.15.19 (Every Element Lies Below the Supremum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $s = \sup A$. Then $x \leq s$ for every $x \in A$.*
Go to proof.

Proposition 0.15.20 (Every Element Lies Above the Infimum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $i = \inf A$. Then $i \leq x$ for every $x \in A$.*

Theorem 0.15.21 (Infimum Is Less Than or Equal to the Supremum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty, bounded below, and bounded above. Then $\inf A \leq \sup A$.*
Go to proof.

Proposition 0.15.22 (Order Comparison of Suprema)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. If every $x \in A$ is below some $y \in B$, then $\sup A \leq \sup B$.*
Go to proof.

Theorem 0.15.23 (Least Upper Bound Property Implies Existence of Suprema)**.** *Let $S \subseteq \mathbb{R}$ be nonempty and bounded above. By the least upper bound property of the real numbers, there exists a real number s such that $s = \sup S$.*

Theorem 0.15.24 (Greatest Lower Bound Property Implies Existence of Infima)**.** *Let $S \subseteq \mathbb{R}$ be nonempty and bounded below. Then there exists a real number i such that $i = \inf S$.*

0.15.3 Maxima and Minima

Definition 0.15.25 (Maximum)**.** Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. An element $m \in S$ is the *maximum* of A if $m \in A$ and every element of A is less than or equal to m.

Definition 0.15.26 (Minimum)**.** Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. An element $m \in S$ is the *minimum* of A if $m \in A$ and every element of A is greater than or equal to m.

Proposition 0.15.27 (Uniqueness of the Maximum)**.** *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $m, n \in S$ are both maxima of A, then $m = n$.*
Go to proof.

Proposition 0.15.28 (Uniqueness of the Minimum)**.** *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $m, n \in S$ are both minima of A, then $m = n$.*

Proposition 0.15.29 (Maximum Implies Supremum). *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $m = \max A$, then $m = \sup A$.*

Go to proof.

Proposition 0.15.30 (Minimum Implies Infimum). *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $m = \min A$, then $m = \inf A$.*

Proposition 0.15.31 (Supremum in the Set is the Maximum). *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $s = \sup A$ and $s \in A$, then $s = \max A$.*

Go to proof.

Proposition 0.15.32 (Infimum in the Set is the Minimum). *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $i = \inf A$ and $i \in A$, then $i = \min A$.*

0.15.4 Epsilon Characterization

ε-Characterization of Supremum

Definition 0.15.33 (Epsilon Characterization of Supremum). Let $A \subseteq \mathbb{R}$ and let $s \in \mathbb{R}$. The element s satisfies the *epsilon characterization of the supremum* of A if and only if s is an upper bound of A and for every $\varepsilon > 0$ there exists $a \in A$ such that $s - \varepsilon < a$.

Definition 0.15.34 (Epsilon Characterization of Infimum). Let $A \subseteq \mathbb{R}$ and let $t \in \mathbb{R}$. The element t satisfies the *epsilon characterization of the infimum* of A if and only if t is a lower bound of A and for every $\varepsilon > 0$ there exists $a \in A$ such that $a < t + \varepsilon$.

0.15.5 Extremal Bounds Summary

0.16 Algebra of Bounds

0.16.1 Set Operations and Bounds

Lemma 0.16.1 (Union Preserves Upper Bounds). *Let $A, B \subseteq \mathbb{R}$. If u_A is an upper bound of A and u_B is an upper bound of B, then $\max\{u_A, u_B\}$ is an upper bound of $A \cup B$. Go to proof.*

Lemma 0.16.2 (Union Preserves Lower Bounds). *Let $A, B \subseteq \mathbb{R}$. If ℓ_A is a lower bound of A and ℓ_B is a lower bound of B, then $\min\{\ell_A, \ell_B\}$ is a lower bound of $A \cup B$. Go to proof.*

Proposition 0.16.3 (Union Is Bounded Above Exactly When Both Pieces Are). *Let $A, B \subseteq \mathbb{R}$ be nonempty. Then $A \cup B$ is bounded above if and only if A is bounded above and B is bounded above. Go to proof.*

Proposition 0.16.4 (Union Is Bounded Below Exactly When Both Pieces Are). *Let $A, B \subseteq \mathbb{R}$ be nonempty. Then $A \cup B$ is bounded below if and only if A is bounded below and B is bounded below. Go to proof.*

Corollary 0.16.5 (Union Is Bounded Exactly When Both Pieces Are). *Let $A, B \subseteq \mathbb{R}$ be nonempty. Then $A \cup B$ is bounded if and only if A is bounded and B is bounded. Go to proof.*

Lemma 0.16.6 (Subsets Preserve Upper Bounds). *Let $C \subseteq A \subseteq \mathbb{R}$. If u is an upper bound of A, then u is an upper bound of C. Go to proof.*

Lemma 0.16.7 (Subsets Preserve Lower Bounds). *Let $C \subseteq A \subseteq \mathbb{R}$. If ℓ is a lower bound of A, then ℓ is a lower bound of C. Go to proof.*

Corollary 0.16.8 (Intersections Inherit Bounds). *Let $A, B \subseteq \mathbb{R}$. Every upper bound of A is an upper bound of $A \cap B$, and every lower bound of A is a lower bound of $A \cap B$. Go to proof.*

Corollary 0.16.9 (Differences Inherit Bounds). *Let $A, B \subseteq \mathbb{R}$. Every upper bound of A is an upper bound of $A \setminus B$, and every lower bound of A is a lower bound of $A \setminus B$. Go to proof.*

Corollary 0.16.10 (Complements Inherit Ambient Bounds). *Let $A \subseteq E \subseteq \mathbb{R}$, and take complements relative to E. Every upper bound of E is an upper bound of $E \setminus A$, and every lower bound of E is a lower bound of $E \setminus A$. Go to proof.*

0.16.2 Lattice Operations and Bounds

Definition 0.16.11 (Pointwise Maximum and Minimum Sets)**.** Let $A, B \subseteq \mathbb{R}$ be nonempty. Define

$$\max(A, B) := \{\max\{a, b\} : a \in A,\ b \in B\},$$

and

$$\min(A, B) := \{\min\{a, b\} : a \in A,\ b \in B\}.$$

Theorem 0.16.12 (Supremum of the Pointwise Maximum Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. Then*

$$\sup \max(A, B) = \max\{\sup A, \sup B\}.$$

Go to proof.

Theorem 0.16.13 (Infimum of the Pointwise Maximum Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded below. Then*

$$\inf \max(A, B) = \max\{\inf A, \inf B\}.$$

Go to proof.

Theorem 0.16.14 (Supremum of the Pointwise Minimum Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. Then*

$$\sup \min(A, B) = \min\{\sup A, \sup B\}.$$

Go to proof.

Theorem 0.16.15 (Infimum of the Pointwise Minimum Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded below. Then*

$$\inf \min(A, B) = \min\{\inf A, \inf B\}.$$

Go to proof.

0.16.3 Images, Inverse Images, and Extrema

Theorem 0.16.16 (Increasing Image Preserves Suprema)**.** *Let $I \subseteq \mathbb{R}$, let $A \subseteq I$ be nonempty and bounded above, and let $f : I \to \mathbb{R}$ be increasing. Suppose $s = \sup A$ belongs to I and f is continuous at s. Then*

$$\sup f(A) = f(\sup A), \qquad f(A) = \{f(a) : a \in A\}.$$

Go to proof.

Theorem 0.16.17 (Increasing Image Preserves Infima)**.** *Let $I \subseteq \mathbb{R}$, let $A \subseteq I$ be nonempty and bounded below, and let $f : I \to \mathbb{R}$ be increasing. Suppose $t = \inf A$ belongs to I and f is continuous at t. Then*

$$\inf f(A) = f(\inf A).$$

Go to proof.

Theorem 0.16.18 (Decreasing Image Sends Infimum to Supremum)**.** *Let $I \subseteq \mathbb{R}$, let $A \subseteq I$ be nonempty and bounded below, and let $f : I \to \mathbb{R}$ be decreasing. Suppose $t = \inf A$ belongs to I and f is continuous at t. Then*

$$\sup f(A) = f(\inf A).$$

Go to proof.

Theorem 0.16.19 (Decreasing Image Sends Supremum to Infimum)**.** *Let $I \subseteq \mathbb{R}$, let $A \subseteq I$ be nonempty and bounded above, and let $f : I \to \mathbb{R}$ be decreasing. Suppose $s = \sup A$ belongs to I and f is continuous at s. Then*

$$\inf f(A) = f(\sup A).$$

Go to proof.

Theorem 0.16.20 (Increasing Inverse Preserves Suprema)**.** *Let $I, J \subseteq \mathbb{R}$, and let $f : I \to J$ be bijective. Suppose $f^{-1} : J \to I$ is increasing and continuous. If $B \subseteq J$ is nonempty and bounded above, and if $\sup B \in J$, then*

$$\sup f^{-1}(B) = f^{-1}(\sup B).$$

Go to proof.

Theorem 0.16.21 (Increasing Inverse Preserves Infima)**.** *Let $I, J \subseteq \mathbb{R}$, and let $f : I \to J$ be bijective. Suppose $f^{-1} : J \to I$ is increasing and continuous. If $B \subseteq J$ is nonempty and bounded below, and if $\inf B \in J$, then*

$$\inf f^{-1}(B) = f^{-1}(\inf B).$$

Go to proof.

Theorem 0.16.22 (Decreasing Inverse Sends Infimum to Supremum)**.** *Let $I, J \subseteq \mathbb{R}$, and let $f : I \to J$ be bijective. Suppose $f^{-1} : J \to I$ is decreasing and continuous. If $B \subseteq J$ is nonempty and bounded below, and if $\inf B \in J$, then*

$$\sup f^{-1}(B) = f^{-1}(\inf B).$$

Go to proof.

Theorem 0.16.23 (Decreasing Inverse Sends Supremum to Infimum)**.** *Let $I, J \subseteq \mathbb{R}$, and let $f : I \to J$ be bijective. Suppose $f^{-1} : J \to I$ is decreasing and continuous. If $B \subseteq J$ is nonempty and bounded above, and if $\sup B \in J$, then*

$$\inf f^{-1}(B) = f^{-1}(\sup B).$$

Go to proof.

0.16.4 Dilation and Displacement

Definition 0.16.24 (Displacement, Dilation, and Reflection)**.** Let $A \subseteq \mathbb{R}$ and let $c, \lambda \in \mathbb{R}$. Define

$$A + c := \{a + c : a \in A\}, \qquad A - c := \{a - c : a \in A\}, \qquad \lambda A := \{\lambda a : a \in A\}.$$

The set $-A := (-1)A$ is the *reflection* of A through the origin.

Proposition 0.16.25 (Translation Preserves Upper Bounds)**.** *Let $A \subseteq \mathbb{R}$ be nonempty. If u is an upper bound of A, then $u + c$ is an upper bound of $A + c$. Go to proof.*

Proposition 0.16.26 (Translation Preserves Lower Bounds)**.** *Let $A \subseteq \mathbb{R}$ be nonempty. If ℓ is a lower bound of A, then $\ell + c$ is a lower bound of $A + c$. Go to proof.*

Proposition 0.16.27 (Positive Dilation Preserves Upper Bounds)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and let $\lambda > 0$. If u is an upper bound of A, then λu is an upper bound of λA. Go to proof.*

Proposition 0.16.28 (Positive Dilation Preserves Lower Bounds)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and let $\lambda > 0$. If ℓ is a lower bound of A, then $\lambda \ell$ is a lower bound of λA. Go to proof.*

Proposition 0.16.29 (Negative Dilation Sends Lower Bounds to Upper Bounds)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and let $\lambda < 0$. If ℓ is a lower bound of A, then $\lambda \ell$ is an upper bound of λA. Go to proof.*

Proposition 0.16.30 (Negative Dilation Sends Upper Bounds to Lower Bounds)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and let $\lambda < 0$. If u is an upper bound of A, then λu is a lower bound of λA. Go to proof.*

Corollary 0.16.31 (Reflection Swaps Upper and Lower Bounds)**.** *Let $A \subseteq \mathbb{R}$ be nonempty. If ℓ is a lower bound of A, then $-\ell$ is an upper bound of $-A$. If u is an upper bound of A, then $-u$ is a lower bound of $-A$. Go to proof.*

0.16.5 Algebra of Suprema and Infima

Current Algebraic Properties of Supremum and Infimum

Definition 0.16.32 (Set Arithmetic Images)**.** Let $A, B \subseteq \mathbb{R}$. Define

$$A + B := \{a + b : a \in A,\ b \in B\}, \qquad A - B := \{a - b : a \in A,\ b \in B\},$$

$$AB := \{ab : a \in A,\ b \in B\}, \qquad A/B := \{a/b : a \in A,\ b \in B,\ b \neq 0\}.$$

Definition 0.16.33 (Reciprocal Image)**.** Let $A \subseteq \mathbb{R}$ and suppose $0 \notin A$. The reciprocal image of A is

$$A^{-1} := \{1/a : a \in A\}.$$

Theorem 0.16.34 (Translation Invariance of the Supremum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $c \in \mathbb{R}$. Then*

$$\sup(A + c) = \sup A + c, \qquad A + c = \{a + c : a \in A\}.$$

Go to proof.

Theorem 0.16.35 (Translation Invariance of the Infimum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $c \in \mathbb{R}$. Then*

$$\inf(A + c) = \inf A + c.$$

Go to proof.

Theorem 0.16.36 (Scalar Multiplication by a Positive Scalar)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $k > 0$. Then*

$$\sup(kA) = k \sup A, \qquad kA = \{ka : a \in A\}.$$

Go to proof.

Theorem 0.16.37 (Positive Scalar Multiplication of the Infimum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $k > 0$. Then*

$$\inf(kA) = k \inf A.$$

Go to proof.

Theorem 0.16.38 (Scalar Multiplication by a Negative Scalar)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $k < 0$. Then*

$$\sup(kA) = k \inf A, \qquad kA = \{ka : a \in A\}.$$

Go to proof.

Theorem 0.16.39 (Negative Scalar Multiplication of the Infimum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $k < 0$. Then*

$$\inf(kA) = k \sup A.$$

Go to proof.

Theorem 0.16.40 (Negation Exchanges Supremum and Infimum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below. Then $-A = \{-a : a \in A\}$ is bounded above and*

$$\sup(-A) = -\inf A.$$

Go to proof.

Theorem 0.16.41 (Supremum of a Sum Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. Then*

$$\sup(A + B) = \sup A + \sup B, \qquad A + B = \{a + b : a \in A,\ b \in B\}.$$

Go to proof.

Theorem 0.16.42 (Infimum of a Sum Set). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded below. Then*

$$\inf(A + B) = \inf A + \inf B.$$

Go to proof.

Theorem 0.16.43 (Supremum of a Difference Set). *Let $A, B \subseteq \mathbb{R}$ be nonempty, with A bounded above and B bounded below. Then*

$$\sup(A - B) = \sup A - \inf B, \qquad A - B = \{a - b : a \in A,\ b \in B\}.$$

Go to proof.

Theorem 0.16.44 (Infimum of a Difference Set). *Let $A, B \subseteq \mathbb{R}$ be nonempty, with A bounded below and B bounded above. Then*

$$\inf(A - B) = \inf A - \sup B.$$

Go to proof.

Theorem 0.16.45 (Supremum of a Dilation). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded, and let $\lambda \in \mathbb{R}$. Then*

$$\sup(\lambda A) = \max\{\lambda \sup A, \lambda \inf A\}, \qquad \lambda A = \{\lambda a : a \in A\}.$$

Go to proof.

Theorem 0.16.46 (Supremum of the Absolute-Value Image). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded. Then*

$$\sup |A| = \max\{|\sup A|, |\inf A|\}, \qquad |A| = \{|a| : a \in A\}.$$

Go to proof.

Theorem 0.16.47 (Supremum of a Reciprocal Set). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded, and suppose that $0 < \inf A$ or $\sup A < 0$. Define*

$$A^{-1} := \{1/a : a \in A\}.$$

Then

$$\sup(A^{-1}) = \frac{1}{\inf A}.$$

Go to proof.

Theorem 0.16.48 (Infimum of a Reciprocal Set). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded, and suppose that $0 < \inf A$ or $\sup A < 0$. Define*

$$A^{-1} := \{1/a : a \in A\}.$$

Then

$$\inf(A^{-1}) = \frac{1}{\sup A}.$$

Go to proof.

Theorem 0.16.49 (Supremum of a Product Set). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded. Then*

$$\sup(AB) = \max\{(\inf A)(\inf B), (\inf A)(\sup B), (\sup A)(\inf B), (\sup A)(\sup B)\}.$$

Go to proof.

Theorem 0.16.50 (Infimum of a Product Set). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded. Then*

$$\inf(AB) = \min\{(\inf A)(\inf B), (\inf A)(\sup B), (\sup A)(\inf B), (\sup A)(\sup B)\}.$$

Go to proof.

Theorem 0.16.51 (Supremum of a Quotient Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded, and suppose $0 < \inf B$ or $\sup B < 0$. Then*

$$\sup(A/B) = \max\left\{\frac{\inf A}{\inf B}, \frac{\inf A}{\sup B}, \frac{\sup A}{\inf B}, \frac{\sup A}{\sup B}\right\}.$$

Go to proof.

Theorem 0.16.52 (Infimum of a Quotient Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded, and suppose $0 < \inf B$ or $\sup B < 0$. Then*

$$\inf(A/B) = \min\left\{\frac{\inf A}{\inf B}, \frac{\inf A}{\sup B}, \frac{\sup A}{\inf B}, \frac{\sup A}{\sup B}\right\}.$$

Go to proof.

Proposition 0.16.53 (Bounds for Suprema and Infima Under Set Addition)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded. Then*

$$\inf A + \inf B \leq \inf(A + B) \qquad \text{and} \qquad \sup(A + B) \leq \sup A + \sup B.$$

Go to proof.

Proposition 0.16.54 (Upper Bounds Depend on the Ambient Order)**.** *Let $(S, \leq_S)$ and $(T, \leq_T)$ be ordered sets, and let $A \subseteq S \cap T$. Whether u is an upper bound of A must be interpreted relative to the chosen ambient order. Go to proof.*

Proposition 0.16.55 (Suprema Depend on the Ambient Set)**.** *Let $A \subseteq S \subseteq S'$ be subsets of an ordered set. The supremum of A relative to S, if it exists, need not agree with the supremum of A relative to S'. Go to proof.*

Theorem 0.16.56 (Ambient Existence of Supremum)**.** *There are ordered sets $S \subseteq S'$ and a subset $A \subseteq S$ such that A has a supremum relative to S' but has no supremum relative to S. Go to proof.*

Lemma 0.16.57 (Rational Gap Example for Suprema)**.** *Let*

$$A = \{q \in \mathbb{Q} : q^2 < 2\}.$$

Then A is nonempty and bounded above in $\mathbb{Q}$, but A has no supremum relative to $\mathbb{Q}$. Go to proof.

Proofs

Capstone

Functions

0.17 Algebra of Functions

Algebra of Functions

Proposition 0.17.1 (Composition of Injections Is Injective)**.** *Let $f : A \to B$ and let $g : B \to C$. If f is injective and g is injective, then $g \circ f : A \to C$ is injective.*

Go to proof.

Proposition 0.17.2 (Composition of Surjections Is Surjective)**.** $\mathrm{Surjective}(f) \wedge \mathrm{Surjective}(g) \Rightarrow \mathrm{Surjective}(g \circ f)$.

Go to proof.

Corollary 0.17.3 (Composition of Bijections Is Bijective)**.** *Let $f : A \to B$ and let $g : B \to C$. If f is bijective and g is bijective, then $g \circ f : A \to C$ is bijective.*

Go to proof.

Proposition 0.17.4 (Inverse of a Bijection Is a Bijection)**.** $\mathrm{Bijective}(f) \Rightarrow \mathrm{Bijective}(f^{-1})$.

Go to proof.

Proposition 0.17.5 (Preimage Distributes over Union and Intersection)**.** *Let $f : A \to B$, $S, T \subseteq B$. Then:*

(i) $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$.

(ii) $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$.

(iii) $f^{-1}(S^c) = (f^{-1}(S))^c$.

Go to proof.

0.18 Real-Valued Functions

Real-Valued Functions

Definition 0.18.1 (Pointwise Sum of Functions)**.** Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. The pointwise sum $f + g : X \to \mathbb{R}$ is the function defined by

$$(f + g)(x) = f(x) + g(x) \qquad (x \in X).$$

Definition 0.18.2 (Pointwise Difference of Functions)**.** Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. The pointwise difference $f - g : X \to \mathbb{R}$ is the function defined by

$$(f - g)(x) = f(x) - g(x) \qquad (x \in X).$$

Definition 0.18.3 (Pointwise Product of Functions)**.** Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. The pointwise product $fg : X \to \mathbb{R}$ is the function defined by

$$(fg)(x) = f(x)g(x) \qquad (x \in X).$$

Definition 0.18.4 (Pointwise Scalar Multiple of a Function)**.** Let $X \subseteq \mathbb{R}$, let $f : X \to \mathbb{R}$, and let $c \in \mathbb{R}$. The pointwise scalar multiple $cf : X \to \mathbb{R}$ is the function defined by

$$(cf)(x) = cf(x) \qquad (x \in X).$$

Definition 0.18.5 (Pointwise Absolute Value of a Function)**.** Let $X \subseteq \mathbb{R}$ and let $f : X \to \mathbb{R}$. The pointwise absolute value $|f| : X \to \mathbb{R}$ is the function defined by

$$|f|(x) = |f(x)| \qquad (x \in X).$$

Definition 0.18.6 (Pointwise Maximum of Two Functions)**.** Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. The pointwise maximum $\max(f, g) : X \to \mathbb{R}$ is the function defined by

$$\max(f, g)(x) = \max\{f(x), g(x)\} \qquad (x \in X).$$

Definition 0.18.7 (Pointwise Quotient of Functions)**.** Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. Suppose $g(x) \neq 0$ for every $x \in X$. The pointwise quotient $f/g : X \to \mathbb{R}$ is the function defined by

$$(f/g)(x) = \frac{f(x)}{g(x)} \qquad (x \in X).$$

Definition 0.18.8 (Linear Combination of Real-Valued Functions)**.** Let $X \subseteq \mathbb{R}$, let $f, g : X \to \mathbb{R}$, and let $\alpha, \beta \in \mathbb{R}$. The real linear combination $\alpha f + \beta g : X \to \mathbb{R}$ is the function defined by

$$(\alpha f + \beta g)(x) = \alpha f(x) + \beta g(x) \qquad (x \in X).$$

Definition 0.18.9 (Real-Linear Rule)**.** Let $\mathcal{C}$ be a class of real-valued functions closed under real linear combinations. A rule T on $\mathcal{C}$ is real-linear if, whenever $f, g \in \mathcal{C}$ and $\alpha, \beta \in \mathbb{R}$,

$$T(\alpha f + \beta g) = \alpha T(f) + \beta T(g)$$

whenever both sides are defined.

Definition 0.18.10 (Bounded Above Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is bounded above on A if there exists $M \in \mathbb{R}$ such that

$$f(x) \leq M \qquad (x \in A).$$

Definition 0.18.11 (Bounded Below Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is bounded below on A if there exists $m \in \mathbb{R}$ such that

$$m \leq f(x) \qquad (x \in A).$$

Definition 0.18.12 (Bounded Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is bounded on A if there exists $B > 0$ such that

$$|f(x)| \leq B \qquad (x \in A).$$

Definition 0.18.13 (Supremum of a Function on a Set)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The supremum of f on A is

$$\sup_{x \in A} f(x) = \sup f(A),$$

provided the supremum of $f(A)$ exists.

Definition 0.18.14 (Infimum of a Function on a Set)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The infimum of f on A is

$$\inf_{x \in A} f(x) = \inf f(A),$$

provided the infimum of $f(A)$ exists.

Definition 0.18.15 (Maximum Point of a Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. A point $x^* \in A$ is a maximum point of f on A if

$$f(x) \leq f(x^*) \qquad (x \in A).$$

Definition 0.18.16 (Minimum Point of a Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. A point $x_* \in A$ is a minimum point of f on A if

$$f(x_*) \leq f(x) \qquad (x \in A).$$

Definition 0.18.17 (Increasing Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is increasing on A if

$$x_1 < x_2 \implies f(x_1) \leq f(x_2) \qquad (x_1, x_2 \in A).$$

Definition 0.18.18 (Strictly Increasing Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is strictly increasing on A if

$$x_1 < x_2 \implies f(x_1) < f(x_2) \qquad (x_1, x_2 \in A).$$

Definition 0.18.19 (Decreasing Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is decreasing on A if

$$x_1 < x_2 \implies f(x_1) \geq f(x_2) \qquad (x_1, x_2 \in A).$$

Definition 0.18.20 (Strictly Decreasing Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is strictly decreasing on A if

$$x_1 < x_2 \implies f(x_1) > f(x_2) \qquad (x_1, x_2 \in A).$$

Definition 0.18.21 (Monotone Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is monotone on A if f is increasing on A or decreasing on A.

Definition 0.18.22 (Constant Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is constant on A if

$$f(x_1) = f(x_2) \qquad (x_1, x_2 \in A).$$

0.19 Subsets of $\mathbb{R}$

Subsets of $\mathbb{R}$

Definition 0.19.1 (ε-Neighbourhood)**.** Let $x \in \mathbb{R}$ and let $\varepsilon > 0$. The **ε-neighbourhood** of x is

$$V_\varepsilon(x) := (x - \varepsilon, x + \varepsilon) = \{y \in \mathbb{R} : |y - x| < \varepsilon\}.$$

Definition 0.19.2 (Deleted ε-Neighbourhood)**.** Let $x \in \mathbb{R}$ and let $\varepsilon > 0$. The **deleted ε-neighbourhood** of x is

$$V_\varepsilon^*(x) := V_\varepsilon(x) \setminus \{x\} = \{y \in \mathbb{R} : 0 < |y - x| < \varepsilon\}.$$

Definition 0.19.3 (Cluster Point)**.** Let $X \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. The point x is a **cluster point** of X if

$$\forall \varepsilon > 0,\ \exists\, y \in X \setminus \{x\} :\ |y - x| < \varepsilon.$$

Theorem 0.19.4 (Sequential Characterization of Cluster Points)**.** *c is a cluster point of A if and only if there exists $(a_n) \subseteq A \setminus \{c\}$ with $a_n \to c$.*

Go to proof.

Definition 0.19.5 (Adherent Point)**.** Let $X \subseteq \mathbb{R}$ and let $x \in \mathbb{R}$. The point x is an **adherent point** of X if

$$\forall \varepsilon > 0 \; \exists y \in X \; (|y - x| < \varepsilon).$$

Definition 0.19.6 (Isolated Point)**.** Let $X \subseteq \mathbb{R}$ and let $x \in X$. The point x is an **isolated point** of X if

$$\exists \varepsilon > 0 \; \big(V_\varepsilon(x) \cap X = \{x\}\big).$$

Definition 0.19.7 (Interior Point)**.** Let $X \subseteq \mathbb{R}$ and let $x \in \mathbb{R}$. The point x is an **interior point** of X if

$$\exists \varepsilon > 0 \; \big(V_\varepsilon(x) \subseteq X\big).$$

Definition 0.19.8 (Boundary Point)**.** Let $X \subseteq \mathbb{R}$ and let $x \in \mathbb{R}$. The point x is a **boundary point** of X if

$$\forall \varepsilon > 0 \; \big(V_\varepsilon(x) \cap X \neq \emptyset\big) \wedge \big(V_\varepsilon(x) \cap X^c \neq \emptyset\big).$$

Definition 0.19.9 (Interior)**.** Let $X \subseteq \mathbb{R}$. The **interior** of X is

$$X^\circ := \{x \in \mathbb{R} : \exists \varepsilon > 0 \; V_\varepsilon(x) \subseteq X\}.$$

Definition 0.19.10 (Boundary)**.** Let $X \subseteq \mathbb{R}$. The **boundary** of X is

$$\partial X := \{x \in \mathbb{R} : \forall \varepsilon > 0 \; (V_\varepsilon(x) \cap X \neq \emptyset \wedge V_\varepsilon(x) \cap X^c \neq \emptyset)\}.$$

Definition 0.19.11 (Closure)**.**

$$\overline{X} := \{x \in \mathbb{R} : \forall \varepsilon > 0 \; V_\varepsilon(x) \cap X \neq \emptyset\}.$$

Lemma 0.19.12 (Elementary Properties of Closures)**.** *Let $X, Y \subseteq \mathbb{R}$. Then: (i) $X \subseteq \overline{X}$; (ii) $\overline{X \cup Y} = \overline{X} \cup \overline{Y}$; (iii) $\overline{X \cap Y} \subseteq \overline{X} \cap \overline{Y}$; (iv) $X \subseteq Y \Rightarrow \overline{X} \subseteq \overline{Y}$.*

Go to proof.

Definition 0.19.13 (Closed Subset of $\mathbb{R}$)**.** E is closed $\iff \overline{E} = E$.

Corollary 0.19.14 (Closed iff Sequentially Closed)**.** *X is closed $\iff \forall (a_n) \subseteq X, \; a_n \to x \Rightarrow x \in X$.*

Go to proof.

Corollary 0.19.15 (Every Point of an Interval Is a Cluster Point)**.** *If I is an interval, then every $x \in I$ is a cluster point of I.*

Go to proof.

Definition 0.19.16 (Bounded Subset of $\mathbb{R}$)**.** X is bounded $\iff \exists M > 0, \; \forall x \in X : |x| \leq M$.

Theorem 0.19.17 (Heine–Borel Theorem for $\mathbb{R}$)**.** *X is closed $\wedge$ X is bounded $\iff$ every $(a_n) \subseteq X$ has a subsequence converging to a limit in X.*

Go to proof.

Definition 0.19.18 (True Near a Point)**.** Property $P(f(x))$ is **true near** x_0 if $\exists \delta > 0, \; \forall x : 0 < |x - x_0| < \delta) \Rightarrow P(f(x))$.

Proofs

Discrete Calculus

0.20 Motivation

Motivation

Discrete Analogs of Calculus

Telescoping Phenomena

Why Discrete Calculus Matters

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0.21 Foundations

0.21.1 Foundations

0.21.2 Summation Notation

Basic Properties of Summation

Splitting Sums

Changing the Index

Empty Sums

Nested Summations

Common Examples

0.21.3 The Binomial Theorem

Theorem 0.21.1 (Binomial Theorem)**.** *For any integer $n \geq 0$ and real numbers x, y,*

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^{k},$$

where the binomial coefficients *are*

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Basic Properties of Binomial Coefficients

Connection with Discrete Calculus

Proposition 0.21.2 (Higher difference formula)**.** *For any function f,*

$$\Delta^n f(x) = \sum_{k=0}^{n} (-1)^{\,n-k} \binom{n}{k} f(x+k).$$

Go to proof.

Explicit Examples

0.22 Algebra of Summations

Linearity

Splitting a Sum

Index Renaming

Index Shifting

Constant Sum

Empty Sum

Nested Sums

Common Summation Identities

0.23 Difference Operators

0.23.1 Difference Operators

Motivation

Forward Difference

Definition 0.23.1 (Forward Difference Operator)**.** Let $f : \mathbb{Z} \to \mathbb{R}$. The *forward difference operator* Δ is defined by

$$\Delta f(n) = f(n+1) - f(n).$$

The function Δf measures the change in f when the input increases by one unit.

Example

0.23.2 Higher Differences

Definition 0.23.2 (Higher Differences)**.** The *second difference* of f is defined by

$$\Delta^2 f(n) = \Delta(\Delta f(n)).$$

More generally, the k-th difference $\Delta^k f(n)$ denotes the k-fold application of the difference operator, defined recursively by

$$\Delta^0 f = f, \qquad \Delta^k f = \Delta(\Delta^{k-1} f).$$

Example

0.23.3 Properties of the Difference Operator

Linearity

Proposition 0.23.3 (Linearity of Δ)**.** *For functions $f, g : \mathbb{Z} \to \mathbb{R}$ and constants $a, b \in \mathbb{R}$,*

$$\Delta(af + bg) = a\,\Delta f + b\,\Delta g.$$

Constant Rule

Proposition 0.23.4 (Constant rule)**.** *If c is a constant function, then $\Delta c = 0$.*

The Shift Operator and Operator Algebra

Definition 0.23.5 (Shift Operator)**.** The *shift operator* E and the *identity operator* I are defined by

$$Ef(n) = f(n+1), \qquad If(n) = f(n).$$

Theorem 0.23.6 ($\Delta = E - I$)**.**

$$\Delta = E - I.$$

0.23.4 Difference Tables and Polynomial Detection

Definition 0.23.7 (Difference Table)**.** Given a sequence $f(0), f(1), f(2), \ldots$, the *difference table* is constructed by listing the successive values of the function and then computing their forward differences $\Delta f(n) = f(n+1) - f(n)$, then $\Delta^2 f$, $\Delta^3 f$, and so on.

Example

Theorem 0.23.8 (Polynomial detection)**.** *Let $f(n)$ be a sequence generated by a polynomial of degree d. Then $\Delta^d f(n)$ is constant, and $\Delta^{d+1} f(n) = 0$.*

0.24 Telescoping Sums

0.24.1 Telescoping Sums

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The Telescoping Identity

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Operator Form

0.25 Discrete Antiderivatives

0.25.1 Discrete Antiderivatives

Definition 0.25.1 (Discrete Antiderivative)**.** Let $f : \mathbb{Z} \to \mathbb{R}$ be a discrete function. A function $F : \mathbb{Z} \to \mathbb{R}$ is called a *discrete antiderivative* (or *discrete indefinite sum*) of f if

$$\Delta F(n) = f(n),$$

that is, $F(n+1) - F(n) = f(n)$ for all $n \in \mathbb{Z}$.

Examples

Constant of Integration

Proposition 0.25.2 (Constant of integration). *If F is a discrete antiderivative of f, then so is $F + C$ for any constant $C \in \mathbb{R}$.*

Fundamental Theorem of Finite Calculus

0.26 Polynomial Basis

0.26.1 Polynomial Basis: Falling Factorials

The Difference Rule for Falling Factorials

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0.27 Calculus Rules

0.27.1 Calculus Rules

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0.28 Expansions

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0.29 Special Functions

0.29.1 Special Functions in Discrete Calculus

The Discrete Exponential

Definition 0.29.1 (Discrete Exponential)**.** For a constant $a \neq -1$ and $n \in \mathbb{Z}$, the *discrete exponential* with base $(1 + a)$ is the function

$$E_a(n) = (1 + a)^n.$$

Proposition 0.29.2 (Discrete exponential rule)**.**

$$\Delta\,(1 + a)^n = a\,(1 + a)^n.$$

0.29.2 The Function 2^n in Discrete Calculus

Differencing 2^n

Antiderivative and Geometric Sum

Higher Differences of 2^n

Proposition 0.29.3 (Higher differences of 2^n). *For all $k \geq 0$,*

$$\Delta^k \, 2^n = 2^n.$$

2^n in Combinatorics and the Newton Expansion

Discrete Differential Equation

Proposition 0.29.4 (Discrete IVP). *The unique function $F : \mathbb{N} \to \mathbb{R}$ satisfying*

$$F(n+1) = 2\,F(n), \qquad F(0) = 1$$

is $F(n) = 2^n$.

Difference Formulas for Special Functions

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Proofs

Sequences

0.30 Sequence Definitions

0.30.1 Definitions

Definition 0.30.1 (Sequence)**.** Let X be a nonempty set. A **sequence** in X is a function

$$x : \mathbb{N} \to X.$$

The value $x(n)$ is written x_n and called the ***n*-th term** of the sequence. The sequence itself is denoted $(x_n)_{n\in\mathbb{N}}$, or simply (x_n) when the index set is clear.

Definition 0.30.2 (Real Sequence)**.** A **real sequence** is a sequence whose image is contained in $\mathbb{R}$. Equivalently, a real sequence is a function

$$x : \mathbb{N} \to \mathbb{R}.$$

0.30.2 Null and Constant Sequences

Null and Constant Sequences

Definition 0.30.3 (Constant Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **constant** exactly when there exists $c \in \mathbb{R}$ such that $x_n = c$ for every $n \in \mathbb{N}$.

Definition 0.30.4 (Null Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is a **null sequence** exactly when for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|x_n| < \varepsilon$ for every $n \geq N$.

Theorem 0.30.5 (Constant Sequence Convergence)**.** *Let (x_n) be a real sequence, and let $c \in \mathbb{R}$. If $x_n = c$ for every $n \in \mathbb{N}$, then (x_n) converges to c.*

Go to proof.

Theorem 0.30.6 (Zero Sequence is Null)**.** *Let (x_n) be a real sequence. If $x_n = 0$ for every $n \in \mathbb{N}$, then (x_n) is a null sequence.*

Go to proof.

Theorem 0.30.7 (Constant Null Sequence)**.** *Let (x_n) be a real sequence, and let $c \in \mathbb{R}$. If $x_n = c$ for every $n \in \mathbb{N}$, then (x_n) is a null sequence if and only if $c = 0$.*

Go to proof.

Definition 0.30.8 (Ultimately Constant Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **ultimately constant** exactly when there exist $N \in \mathbb{N}$ and $c \in \mathbb{R}$ such that $x_n = c$ for every $n \geq N$.

Theorem 0.30.9 (Difference from the Limit is Null)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. The sequence (x_n) converges to L if and only if the sequence $(x_n - L)$ is a null sequence.*

Go to proof.

Theorem 0.30.10 (Ultimately Constant Sequence Convergence)**.** *Let (x_n) be a real sequence. If there exist $N_0 \in \mathbb{N}$ and $c \in \mathbb{R}$ such that $x_n = c$ for every $n \geq N_0$, then (x_n) converges to c.*

Go to proof.

Theorem 0.30.11 (Constant Implies Ultimately Constant)**.** *Every constant real sequence is ultimately constant.*

Go to proof.

Theorem 0.30.12 (Ultimately Zero Sequence is Null)**.** *Let (x_n) be a real sequence. If there exists $N_0 \in \mathbb{N}$ such that $x_n = 0$ for every $n \geq N_0$, then (x_n) is a null sequence.*

Go to proof.

Theorem 0.30.13 (Ultimately Constant Null Sequence)**.** *Let (x_n) be a real sequence. Suppose there exist $N_0 \in \mathbb{N}$ and $c \in \mathbb{R}$ such that $x_n = c$ for every $n \geq N_0$. Then (x_n) is a null sequence if and only if $c = 0$.*

Go to proof.

Theorem 0.30.14 (Tail Equality Preserves Convergence)**.** *Let (x_n) and (y_n) be real sequences. If there exists $N_0 \in \mathbb{N}$ such that $x_n = y_n$ for every $n \geq N_0$, then for every $L \in \mathbb{R}$, (x_n) converges to L if and only if (y_n) converges to L.*

Go to proof.

Definition 0.30.15 (Sequence Bounded Above)**.** Let (x_n) be a real sequence. The sequence (x_n) is **bounded above** exactly when there exists $M \in \mathbb{R}$ such that $x_n \leq M$ for every $n \in \mathbb{N}$.

Definition 0.30.16 (Sequence Bounded Below)**.** Let (x_n) be a real sequence. The sequence (x_n) is **bounded below** exactly when there exists $m \in \mathbb{R}$ such that $m \leq x_n$ for every $n \in \mathbb{N}$.

Definition 0.30.17 (Bounded Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **bounded** exactly when there exists $M > 0$ such that $|x_n| \leq M$ for every $n \in \mathbb{N}$.

Theorem 0.30.18 (Eventually Bounded Above Tail Formulation)**.** *Let (x_n) be a real sequence. The sequence (x_n) is bounded above if and only if there exist $N_0 \in \mathbb{N}$ and $M \in \mathbb{R}$ such that $x_n \leq M$ for every $n \geq N_0$.*

Go to proof.

Theorem 0.30.19 (Eventually Bounded Below Tail Formulation)**.** *Let (x_n) be a real sequence. The sequence (x_n) is bounded below if and only if there exist $N_0 \in \mathbb{N}$ and $m \in \mathbb{R}$ such that $m \leq x_n$ for every $n \geq N_0$.*

Go to proof.

Theorem 0.30.20 (Eventually Bounded Tail Formulation)**.** *Let (x_n) be a real sequence. The sequence (x_n) is bounded if and only if there exist $N_0 \in \mathbb{N}$ and $M > 0$ such that $|x_n| \leq M$ for every $n \geq N_0$.*

Go to proof.

Theorem 0.30.21 (Bounded Sequence If and Only If Bounded Above and Below)**.** *Let (x_n) be a real sequence. The sequence (x_n) is bounded if and only if it is both bounded above and bounded below.*

Go to proof.

Theorem 0.30.22 (Absolute Bound Gives Upper and Lower Bounds)**.** *Let (x_n) be a real sequence, and let $K > 0$. If $|x_n| \leq K$ for every $n \in \mathbb{N}$, then $-K \leq x_n \leq K$ for every $n \in \mathbb{N}$.*

Go to proof.

Theorem 0.30.23 (Upper and Lower Bounds Give an Absolute Bound)**.** *Let (x_n) be a real sequence. If there exist $m, M \in \mathbb{R}$ such that $m \leq x_n \leq M$ for every $n \in \mathbb{N}$, then there exists $K > 0$ such that $|x_n| \leq K$ for every $n \in \mathbb{N}$.*

Go to proof.

0.31 Examples and Counterexamples

Examples and Counterexamples

0.32 Convergence

Definition 0.32.1 (Convergence of a Sequence)**.** Let $(x_n) \subseteq \mathbb{R}$. The sequence (x_n) **converges** to $L \in \mathbb{R}$ if

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(|x_n - L| < \varepsilon\big).$$

The value L is called the **limit** of (x_n), written $x_n \to L$ or $\lim_{n\to\infty} x_n = L$. A sequence that converges to some $L \in \mathbb{R}$ is called **convergent**; otherwise it is called **divergent**.

Definition 0.32.2 (Convergence — Neighborhood Formulation)**.** Let $(x_n) \subseteq \mathbb{R}$ and let $L \in \mathbb{R}$. The sequence (x_n) **converges** to L if for every $\varepsilon > 0$ the ε-neighborhood

$$V_\varepsilon(L) \ := \ (L - \varepsilon,\, L + \varepsilon)$$

contains all but finitely many terms of (x_n); that is,

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(x_n \in V_\varepsilon(L)\big).$$

Theorem 0.32.3 (Equivalence of Convergence Formulations)**.** *Let $(x_n) \subseteq \mathbb{R}$ and let $L \in \mathbb{R}$. The following statements are equivalent.*

(a) $x_n \to L$.

(b) $\forall \varepsilon > 0\ \exists K \in \mathbb{N}\ \forall n \geq K\ \big(|x_n - L| < \varepsilon\big)$.

(c) $\forall \varepsilon > 0\ \exists K \in \mathbb{N}\ \forall n \geq K\ \big(L - \varepsilon < x_n < L + \varepsilon\big)$.

(d) $\forall \varepsilon > 0\ \exists K \in \mathbb{N}\ \forall n \geq K\ \big(x_n \in V_\varepsilon(L)\big)$.

Go to proof.

0.32.1 Limits

Theorem 0.32.4 (Uniqueness of Limits)**.** *Let $(x_n) \subseteq \mathbb{R}$. If $x_n \to L$ and $x_n \to K$, then $L = K$.*

Go to proof.

Theorem 0.32.5 (Limit Preserves Eventual Order)**.** *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. Suppose that $x_n \to L$ and $y_n \to M$. If there exists $N_0 \in \mathbb{N}$ such that $x_n \leq y_n$ for every $n \geq N_0$, then $L \leq M$.*

Go to proof.

Theorem 0.32.6 (Strict Limit Separation Gives Eventual Order)**.** *Let (x_n) and (y_n) be real sequences, and let $A, B \in \mathbb{R}$. Suppose that $x_n \to A$ and $y_n \to B$. If $A < B$, then there exists $N \in \mathbb{N}$ such that $x_n < y_n$ for every $n \geq N$.*

Go to proof.

Theorem 0.32.7 (Eventual Strict Comparison Preserves Weak Limit Order)**.** *Let (x_n) and (y_n) be real sequences, and let $A, B \in \mathbb{R}$. Suppose that $x_n \to A$ and $y_n \to B$.*

(a) If there exists $N \in \mathbb{N}$ such that $x_n < y_n$ for every $n \geq N$, then $A \leq B$.

(b) If there exists $N \in \mathbb{N}$ such that $x_n > y_n$ for every $n \geq N$, then $A \geq B$.

Go to proof.

Theorem 0.32.8 (Constant Comparison for Sequence Limits)**.** *Let (x_n) be a real sequence, and let $A, B \in \mathbb{R}$. Suppose that $x_n \to A$.*

(a) If there exists $N \in \mathbb{N}$ such that $x_n \leq B$ for every $n \geq N$, then $A \leq B$.

(b) If there exists $N \in \mathbb{N}$ such that $x_n < B$ for every $n \geq N$, then $A \leq B$.

(c) If there exists $N \in \mathbb{N}$ such that $x_n \geq B$ for every $n \geq N$, then $A \geq B$.

(d) If there exists $N \in \mathbb{N}$ such that $x_n > B$ for every $n \geq N$, then $A \geq B$.

Go to proof.

Theorem 0.32.9 (Constant Squeeze Theorem)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If there exists $N_0 \in \mathbb{N}$ such that $L \leq x_n \leq L$ for every $n \geq N_0$, then $x_n \to L$.*

Go to proof.

Theorem 0.32.10 (Sequence Squeeze Theorem)**.** *Let (a_n), (x_n), and (b_n) be real sequences, and let $L \in \mathbb{R}$. Suppose that $a_n \to L$ and $b_n \to L$. If there exists $N_0 \in \mathbb{N}$ such that $a_n \leq x_n \leq b_n$ for every $n \geq N_0$, then $x_n \to L$.*

Go to proof.

Theorem 0.32.11 (Absolute-Value Squeeze Theorem)**.** *Let (x_n) and (u_n) be real sequences, and let $L \in \mathbb{R}$. Suppose that $u_n \to 0$. If there exists $N_0 \in \mathbb{N}$ such that $|x_n - L| \leq u_n$ for every $n \geq N_0$, then $x_n \to L$.*

Go to proof.

0.32.2 Algebra of Limits

Algebra of Convergent Sequences

Definition 0.32.12 (Pointwise Sum of Sequences)**.** Let (x_n) and (y_n) be real sequences. The **pointwise sum** of (x_n) and (y_n) is the real sequence $(x_n + y_n)$ defined by

$$(x + y)_n := x_n + y_n \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.13 (Pointwise Difference of Sequences)**.** Let (x_n) and (y_n) be real sequences. The **pointwise difference** of (x_n) and (y_n) is the real sequence $(x_n - y_n)$ defined by

$$(x - y)_n := x_n - y_n \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.14 (Scalar Multiple of a Sequence)**.** Let (x_n) be a real sequence, and let $\alpha \in \mathbb{R}$. The **scalar multiple** $\alpha(x_n)$ is the real sequence (αx_n) defined by

$$(\alpha x)_n := \alpha x_n \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.15 (Linear Combination of Real Sequences)**.** Let (x_n) and (y_n) be real sequences, and let $\alpha, \beta \in \mathbb{R}$. The real linear combination $\alpha(x_n) + \beta(y_n)$ is the real sequence $(\alpha x_n + \beta y_n)$ defined by

$$(\alpha x + \beta y)_n := \alpha x_n + \beta y_n \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.16 (Pointwise Negation of a Sequence)**.** Let (x_n) be a real sequence. The **pointwise negation** of (x_n) is the real sequence $(-x_n)$ defined by

$$(-x)_n := -x_n \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.17 (Pointwise Product of Sequences)**.** Let (x_n) and (y_n) be real sequences. The **pointwise product** of (x_n) and (y_n) is the real sequence $(x_n y_n)$ defined by

$$(xy)_n := x_n y_n \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.18 (Reciprocal Sequence)**.** Let (x_n) be a real sequence such that $x_n \neq 0$ for every $n \in \mathbb{N}$. The **reciprocal sequence** of (x_n) is the real sequence $(1/x_n)$ defined by

$$\left(\frac{1}{x}\right)_n := \frac{1}{x_n} \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.19 (Pointwise Quotient of Sequences)**.** Let (x_n) and (y_n) be real sequences such that $y_n \neq 0$ for every $n \in \mathbb{N}$. The **pointwise quotient** of (x_n) by (y_n) is the real sequence (x_n/y_n) defined by

$$\left(\frac{x}{y}\right)_n := \frac{x_n}{y_n} \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.20 (Square Sequence)**.** Let (x_n) be a real sequence. The **square sequence** of (x_n) is the real sequence (x_n^2) defined by

$$(x^2)_n := x_n^2 \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.21 (Absolute Value Sequence)**.** Let (x_n) be a real sequence. The **absolute value sequence** of (x_n) is the real sequence $(|x_n|)$ defined by

$$(|x|)_n := |x_n| \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.22 (Square Root Sequence)**.** Let (x_n) be a real sequence such that $0 \leq x_n$ for every $n \in \mathbb{N}$. The **square root sequence** of (x_n) is the real sequence $(\sqrt{x_n})$ defined by

$$(\sqrt{x})_n := \sqrt{x_n} \qquad \text{for every } n \in \mathbb{N}.$$

Theorem 0.32.23 (Limit of a Scalar Multiple)**.** *Let (x_n) be a real sequence, let $L \in \mathbb{R}$, and let $\alpha \in \mathbb{R}$. If $x_n \to L$, then $\alpha x_n \to \alpha L$.*

Go to proof.

Theorem 0.32.24 (Limit of a Sum)**.** *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. If $x_n \to L$ and $y_n \to M$, then $x_n + y_n \to L + M$.*

Go to proof.

Theorem 0.32.25 (Limit of a Negation)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If $x_n \to L$, then $-x_n \to -L$.*

Go to proof.

Theorem 0.32.26 (Limit of a Difference)**.** *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. If $x_n \to L$ and $y_n \to M$, then $x_n - y_n \to L - M$.*

Go to proof.

Theorem 0.32.27 (Limit of a Product)**.** *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. If $x_n \to L$ and $y_n \to M$, then $x_n y_n \to LM$.*

Go to proof.

Theorem 0.32.28 (Nonzero Limit is Eventually Nonzero)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$ with $L \neq 0$. If $x_n \to L$, then there exists $N \in \mathbb{N}$ such that $x_n \neq 0$ for every $n \geq N$.*

Go to proof.

Theorem 0.32.29 (Limit of a Reciprocal)**.** *Let (x_n) be a real sequence such that $x_n \neq 0$ for every $n \in \mathbb{N}$, and let $L \in \mathbb{R}$ with $L \neq 0$. If $x_n \to L$, then $1/x_n \to 1/L$.*

Go to proof.

Theorem 0.32.30 (Limit of a Quotient)**.** *Let (x_n) and (y_n) be real sequences such that $y_n \neq 0$ for every $n \in \mathbb{N}$, and let $L, M \in \mathbb{R}$ with $M \neq 0$. If $x_n \to L$ and $y_n \to M$, then $x_n/y_n \to L/M$.*

Go to proof.

Theorem 0.32.31 (Limit of a Square)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If $x_n \to L$, then $x_n^2 \to L^2$.*

Go to proof.

Theorem 0.32.32 (Limit of an Absolute Value)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If $x_n \to L$, then $|x_n| \to |L|$.*

Go to proof.

Theorem 0.32.33 (Positive Limit is Eventually Positive)**.** *Let (x_n) be a real sequence, and let $L > 0$. If $x_n \to L$, then there exists $N \in \mathbb{N}$ such that $0 < x_n$ for every $n \geq N$.*

Go to proof.

Theorem 0.32.34 (Limit of a Square Root)**.** *Let (x_n) be a real sequence such that $0 \leq x_n$ for every $n \in \mathbb{N}$, and let $L \in \mathbb{R}$. If $x_n \to L$, then $0 \leq L$ and $\sqrt{x_n} \to \sqrt{L}$.*

Go to proof.

Theorem 0.32.35 (Polynomial Sequence Limit)**.** *Let (x_n) be a real sequence, let $L \in \mathbb{R}$, and let $p \in \mathbb{R}[t]$ be a polynomial. If $x_n \to L$, then $p(x_n) \to p(L)$.*

Go to proof.

Theorem 0.32.36 (Rational Sequence Limit)**.** *Let (x_n) be a real sequence, let $L \in \mathbb{R}$, and let $p, q \in \mathbb{R}[t]$ be polynomials. Suppose $q(L) \neq 0$ and $q(x_n) \neq 0$ for every $n \in \mathbb{N}$. If $x_n \to L$, then*

$$\frac{p(x_n)}{q(x_n)} \to \frac{p(L)}{q(L)}.$$

Go to proof.

0.33 Tails

Definition 0.33.1 (M-Tail of a Sequence)**.** Let $(x_n) \subseteq \mathbb{R}$ and let $M \in \mathbb{N}$. The **M-tail** of (x_n) is the sequence

$$(x_n)_{n \geq M} \;:=\; \big(x_M, \, x_{M+1}, \, x_{M+2}, \, \ldots\big).$$

Theorem 0.33.2 (Convergence of a Tail)**.** *Let $(x_n) \subseteq \mathbb{R}$ and let $m \in \mathbb{N}$. The m-tail $(x_{m+n})_{n \in \mathbb{N}}$ converges if and only if (x_n) converges. Moreover, in this case,*

$$\lim_{n \to \infty} x_{m+n} \;=\; \lim_{n \to \infty} x_n.$$

Go to proof.

Theorem 0.33.3 (Convergence by Domination)**.** *Let $(x_n) \subseteq \mathbb{R}$ and let $L \in \mathbb{R}$. Let (a_n) be a sequence of positive real numbers with $\lim\limits_{n \to \infty} a_n = 0$. If there exist a constant $c > 0$ and an index $m \in \mathbb{N}$ such that*

$$|x_n - L| \;\leq\; c\, a_n \qquad \forall n \geq m,$$

then $\lim\limits_{n \to \infty} x_n = L$.

Go to proof.

Theorem 0.33.4 (Ratio Limit Less Than One Implies Null)**.** *Let (x_n) be a sequence of positive real numbers. Suppose that the limit*

$$L := \lim_{n \to \infty} \frac{x_{n+1}}{x_n}$$

exists. If $L < 1$, then (x_n) converges and

$$\lim_{n \to \infty} x_n = 0.$$

Go to proof.

0.34 Monotonicity

Monotonicity of Sequences

Definition 0.34.1 (Increasing Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **increasing** if

$$x_n \le x_{n+1} \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.34.2 (Decreasing Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **decreasing** if

$$x_{n+1} \le x_n \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.34.3 (Strictly Increasing Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **strictly increasing** if

$$x_n < x_{n+1} \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.34.4 (Strictly Decreasing Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **strictly decreasing** if

$$x_{n+1} < x_n \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.34.5 (Monotone Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **monotone** if it is increasing or decreasing.

Definition 0.34.6 (Eventually Increasing Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **eventually increasing** if there exists $N \in \mathbb{N}$ such that

$$x_n \le x_{n+1} \qquad \text{for every } n \ge N.$$

Definition 0.34.7 (Eventually Decreasing Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **eventually decreasing** if there exists $N \in \mathbb{N}$ such that

$$x_{n+1} \le x_n \qquad \text{for every } n \ge N.$$

Definition 0.34.8 (Eventually Monotone Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **eventually monotone** if it is eventually increasing or eventually decreasing.

Theorem 0.34.9 (Monotone Convergence Theorem)**.** *Let (x_n) be a real sequence.*

(a) If (x_n) is increasing and bounded above, then (x_n) converges and

$$\lim_{n\to\infty} x_n = \sup\{x_n : n \in \mathbb{N}\}.$$

(b) If (x_n) is decreasing and bounded below, then (x_n) converges and

$$\lim_{n\to\infty} x_n = \inf\{x_n : n \in \mathbb{N}\}.$$

Go to proof.

Theorem 0.34.10 (Strict Monotonicity Implies Monotonicity)**.** *Let (x_n) be a real sequence.*

(a) If (x_n) is strictly increasing, then (x_n) is increasing.

(b) If (x_n) is strictly decreasing, then (x_n) is decreasing.

Go to proof.

Theorem 0.34.11 (Bounded Monotone Sequence Equivalences)**.** *Let (x_n) be a real sequence.*

(a) If (x_n) is increasing, then (x_n) converges if and only if (x_n) is bounded above.

(b) If (x_n) is decreasing, then (x_n) converges if and only if (x_n) is bounded below.

(c) If (x_n) is monotone, then (x_n) converges if and only if (x_n) is bounded.

Go to proof.

Theorem 0.34.12 (Eventually Monotone Convergence Theorem)**.** *Let (x_n) be a real sequence.*

(a) If (x_n) is eventually increasing and bounded above, then (x_n) converges.

(b) If (x_n) is eventually decreasing and bounded below, then (x_n) converges.

(c) If (x_n) is eventually monotone and bounded, then (x_n) converges.

Go to proof.

Theorem 0.34.13 (Unbounded Monotone Divergence)**.** *Let (x_n) be a real sequence.*

(a) If (x_n) is increasing and is not bounded above, then $x_n \to +\infty$.

(b) If (x_n) is decreasing and is not bounded below, then $x_n \to -\infty$.

Go to proof.

Theorem 0.34.14 (Algebraic Transformations Preserve Monotonicity)**.** *Let (x_n) be a real sequence and let $c, \alpha \in \mathbb{R}$.*

(a) If (x_n) is increasing, then $(x_n + c)$ is increasing.

(b) If (x_n) is decreasing, then $(x_n + c)$ is decreasing.

(c) If (x_n) is increasing and $\alpha > 0$, then (αx_n) is increasing.

(d) If (x_n) is increasing and $\alpha < 0$, then (αx_n) is decreasing.

(e) If (x_n) is increasing, then $(-x_n)$ is decreasing.

(f) If (x_n) is decreasing, then $(-x_n)$ is increasing.

Go to proof.

0.35 Subsequences

Subsequences

Definition 0.35.1 (Strictly Increasing Index Map)**.** Let $\sigma : \mathbb{N} \to \mathbb{N}$ be a function. The function σ is a **strictly increasing index map** if

$$k < \ell \Longrightarrow \sigma(k) < \sigma(\ell) \qquad \text{for all } k, \ell \in \mathbb{N}.$$

Definition 0.35.2 (Subsequence)**.** Let (x_n) be a real sequence. A **subsequence** of (x_n) is a sequence $(x_{\sigma(k)})_{k\in\mathbb{N}}$, where $\sigma : \mathbb{N} \to \mathbb{N}$ is a strictly increasing index map.

Definition 0.35.3 (Subsequential Limit)**.** Let (x_n) be a real sequence and let $L \in \mathbb{R}$. The real number L is a **subsequential limit** of (x_n) if there exists a subsequence $(x_{\sigma(k)})$ such that

$$\lim_{k\to\infty} x_{\sigma(k)} = L.$$

Definition 0.35.4 (Has Convergent Subsequence)**.** Let (x_n) be a real sequence. The sequence (x_n) **has a convergent subsequence** if there exist a strictly increasing index map $\sigma : \mathbb{N} \to \mathbb{N}$ and a real number L such that

$$\lim_{k\to\infty} x_{\sigma(k)} = L.$$

Theorem 0.35.5 (Subsequence Indices Dominate the Identity)**.** *Let $\sigma : \mathbb{N} \to \mathbb{N}$ be a strictly increasing index map. Then*

$$k \le \sigma(k) \qquad \text{for every } k \in \mathbb{N}.$$

Go to proof.

Theorem 0.35.6 (Subsequences Preserve Limits)**.** *Let (x_n) be a real sequence and let $L \in \mathbb{R}$. If $x_n \to L$, then every subsequence of (x_n) converges to L.*

Go to proof.

Theorem 0.35.7 (Subsequential Limit of a Convergent Sequence)**.** *Let (x_n) be a real sequence and let $L \in \mathbb{R}$. If $x_n \to L$, then L is the only possible subsequential limit of (x_n).*

Go to proof.

Theorem 0.35.8 (Divergence by Two Subsequential Limits)**.** *Let (x_n) be a real sequence. If (x_n) has two distinct subsequential limits, then (x_n) does not converge.*

Go to proof.

Theorem 0.35.9 (Boundedness Passes to Subsequences)**.** *Every subsequence of a bounded real sequence is bounded.*

Go to proof.

Theorem 0.35.10 (Monotonicity Passes to Subsequences)**.** *Every subsequence of an increasing real sequence is increasing, and every subsequence of a decreasing real sequence is decreasing.*

Go to proof.

Theorem 0.35.11 (Subsequence of a Subsequence)**.** *Every subsequence of a subsequence of (x_n) is a subsequence of (x_n).*

Go to proof.

Theorem 0.35.12 (Eventual Properties Pass to Subsequences)**.** *Let $P(n)$ be a property of indices. If there exists $N \in \mathbb{N}$ such that $P(n)$ holds for every $n \ge N$, then for every strictly increasing index map σ there exists $K \in \mathbb{N}$ such that $P(\sigma(k))$ holds for every $k \ge K$.*

Go to proof.

Theorem 0.35.13 (Frequent Properties Yield Subsequences)**.** *Let $P(n)$ be a property of indices. If for every $N \in \mathbb{N}$ there exists $n \ge N$ such that $P(n)$ holds, then there exists a strictly increasing index map $\sigma : \mathbb{N} \to \mathbb{N}$ such that $P(\sigma(k))$ holds for every $k \in \mathbb{N}$.*

Go to proof.

Theorem 0.35.14 (Subsequential Limits Respect Bounds)**.** *Let (x_n) be a real sequence and let L be a subsequential limit of (x_n). If $m \le x_n \le M$ for every $n \in \mathbb{N}$, then $m \le L \le M$.*

Go to proof.

Theorem 0.35.15 (Squeeze Passes to Subsequences)**.** *Let (a_n), (x_n), and (b_n) be real sequences. If $a_n \le x_n \le b_n$ eventually, then for every subsequence $(x_{\sigma(k)})$ there exists $K \in \mathbb{N}$ such that*

$$a_{\sigma(k)} \le x_{\sigma(k)} \le b_{\sigma(k)} \qquad \text{for every } k \ge K.$$

Go to proof.

Theorem 0.35.16 (Monotone Subsequence Theorem)**.** *Every real sequence has a monotone subsequence.*

Go to proof.

Theorem 0.35.17 (Bolzano-Weierstrass Theorem for Sequences)**.** *Every bounded real sequence has a convergent subsequence.*

Go to proof.

Theorem 0.35.18 (Sequential Compactness of Closed Bounded Intervals)**.** *Let $a, b \in \mathbb{R}$ with $a \leq b$. Every sequence in $[a, b]$ has a convergent subsequence whose limit belongs to $[a, b]$.*

Go to proof.

0.36 Divergence

Divergence

Definition 0.36.1 (Divergent Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **divergent** if there is no $L \in \mathbb{R}$ such that $x_n \to L$.

Definition 0.36.2 (Divergence to Positive Infinity)**.** Let (x_n) be a real sequence. The sequence (x_n) **diverges to** $+\infty$ if

$$\forall M \in \mathbb{R}\, \exists N \in \mathbb{N}\, \forall n \geq N\, \big(M < x_n\big).$$

Definition 0.36.3 (Divergence to Negative Infinity)**.** Let (x_n) be a real sequence. The sequence (x_n) **diverges to** $-\infty$ if

$$\forall M \in \mathbb{R}\, \exists N \in \mathbb{N}\, \forall n \geq N\, \big(x_n < M\big).$$

Definition 0.36.4 (Oscillatory Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **oscillatory** if it is divergent, does not diverge to $+\infty$, and does not diverge to $-\infty$.

Theorem 0.36.5 (Divergence to Infinity Implies Real Divergence)**.** *Let (x_n) be a real sequence. If $x_n \to +\infty$ or $x_n \to -\infty$, then (x_n) is divergent.*

Go to proof.

Theorem 0.36.6 (Two Subsequential Limits Force Divergence)**.** *Let (x_n) be a real sequence. If (x_n) has two subsequential limits $L, K \in \mathbb{R}$ with $L \neq K$, then (x_n) is divergent.*

Go to proof.

Theorem 0.36.7 (Unbounded Above Sequence Has a Positive-Infinity Subsequence)**.** *Let (x_n) be a real sequence. If (x_n) is not bounded above, then there exists a subsequence $(x_{\sigma(k)})$ such that $x_{\sigma(k)} \to +\infty$.*

Go to proof.

Theorem 0.36.8 (Unbounded Below Sequence Has a Negative-Infinity Subsequence)**.** *Let (x_n) be a real sequence. If (x_n) is not bounded below, then there exists a subsequence $(x_{\sigma(k)})$ such that $x_{\sigma(k)} \to -\infty$.*

Go to proof.

Theorem 0.36.9 (Bounded Divergence Produces Two Subsequential Limits)**.** *Let (x_n) be a bounded real sequence. If (x_n) is divergent, then there exist $L, K \in \mathbb{R}$ with $L \neq K$ such that L and K are subsequential limits of (x_n).*

Go to proof.

0.37 Liminf and Limsup

Liminf and Limsup

Definition 0.37.1 (Tail Supremum Sequence)**.** Let (x_n) be a bounded real sequence. The **tail supremum sequence** of (x_n) is the real sequence (s_n) defined by

$$s_n := \sup\{x_k : k \geq n\} \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.37.2 (Tail Infimum Sequence)**.** Let (x_n) be a bounded real sequence. The **tail infimum sequence** of (x_n) is the real sequence (i_n) defined by

$$i_n := \inf\{x_k : k \geq n\} \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.37.3 (Limit Superior of a Sequence)**.** Let (x_n) be a bounded real sequence, and let (s_n) be its tail supremum sequence. The **limit superior** of (x_n) is

$$\limsup_{n\to\infty} x_n := \lim_{n\to\infty} s_n.$$

Definition 0.37.4 (Limit Inferior of a Sequence)**.** Let (x_n) be a bounded real sequence, and let (i_n) be its tail infimum sequence. The **limit inferior** of (x_n) is

$$\liminf_{n\to\infty} x_n := \lim_{n\to\infty} i_n.$$

Theorem 0.37.5 (Tail Suprema Are Decreasing)**.** *Let (x_n) be a bounded real sequence, and let (s_n) be its tail supremum sequence. Then (s_n) is decreasing.*

Go to proof.

Theorem 0.37.6 (Tail Infima Are Increasing)**.** *Let (x_n) be a bounded real sequence, and let (i_n) be its tail infimum sequence. Then (i_n) is increasing.*

Go to proof.

Theorem 0.37.7 (Liminf Is Below Limsup)**.** *Let (x_n) be a bounded real sequence. Then*

$$\liminf_{n\to\infty} x_n \leq \limsup_{n\to\infty} x_n.$$

Go to proof.

Theorem 0.37.8 (Convergence iff Liminf Equals Limsup)**.** *Let (x_n) be a bounded real sequence, and let $L \in \mathbb{R}$. Then $x_n \to L$ if and only if*

$$\liminf_{n\to\infty} x_n = \limsup_{n\to\infty} x_n = L.$$

Go to proof.

Theorem 0.37.9 (Limsup Is the Largest Subsequential Limit)**.** *Let (x_n) be a bounded real sequence, and let $S = \limsup_{n\to\infty} x_n$. Then S is a subsequential limit of (x_n), and every subsequential limit of (x_n) is less than or equal to S.*

Go to proof.

Theorem 0.37.10 (Liminf Is the Smallest Subsequential Limit)**.** *Let (x_n) be a bounded real sequence, and let $I = \liminf_{n\to\infty} x_n$. Then I is a subsequential limit of (x_n), and every subsequential limit of (x_n) is greater than or equal to I.*

Go to proof.

Theorem 0.37.11 (Oscillation Criterion via Liminf and Limsup). *Let (x_n) be a bounded real sequence. Then*

$$\liminf_{n\to\infty} x_n < \limsup_{n\to\infty} x_n$$

if and only if (x_n) has two distinct subsequential limits.

Go to proof.

Theorem 0.37.12 (Limsup Comparison Under Eventual Order). *Let (x_n) and (y_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that $x_n \le y_n$ for every $n \ge N$, then*

$$\limsup_{n\to\infty} x_n \le \limsup_{n\to\infty} y_n.$$

Go to proof.

Theorem 0.37.13 (Liminf Comparison Under Eventual Order). *Let (x_n) and (y_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that $x_n \le y_n$ for every $n \ge N$, then*

$$\liminf_{n\to\infty} x_n \le \liminf_{n\to\infty} y_n.$$

Go to proof.

Theorem 0.37.14 (Limsup Squeeze Under Eventual Order). *Let (a_n), (x_n), and (b_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that*

$$a_n \le x_n \le b_n \qquad \text{for every } n \ge N,$$

then

$$\limsup_{n\to\infty} a_n \le \limsup_{n\to\infty} x_n \le \limsup_{n\to\infty} b_n.$$

Go to proof.

Theorem 0.37.15 (Liminf Squeeze Under Eventual Order). *Let (a_n), (x_n), and (b_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that*

$$a_n \le x_n \le b_n \qquad \text{for every } n \ge N,$$

then

$$\liminf_{n\to\infty} a_n \le \liminf_{n\to\infty} x_n \le \liminf_{n\to\infty} b_n.$$

Go to proof.

0.38 Order-Theoretic Limit Constructions

Order-Theoretic Limit Constructions

Theorem 0.38.1 (Increasing Sequence Limit as Supremum). *Let (x_n) be an increasing real sequence that is bounded above. Then*

$$\lim_{n\to\infty} x_n = \sup\{x_n : n \in \mathbb{N}\}.$$

Go to proof.

Theorem 0.38.2 (Decreasing Sequence Limit as Infimum). *Let (x_n) be a decreasing real sequence that is bounded below. Then*

$$\lim_{n\to\infty} x_n = \inf\{x_n : n \in \mathbb{N}\}.$$

Go to proof.

Theorem 0.38.3 (Tail Suprema and Infima Converge)**.** *Let (x_n) be a bounded real sequence. Define*

$$s_n = \sup\{x_k : k \geq n\}, \qquad i_n = \inf\{x_k : k \geq n\}.$$

Then (s_n) and (i_n) converge.

Go to proof.

Theorem 0.38.4 (Bounded Sequence Has Limsup and Liminf)**.** *Every bounded real sequence has a limit superior and a limit inferior.*

Go to proof.

0.39 Cluster Values of Sequences

Cluster Values of Sequences

Definition 0.39.1 (Cluster Value of a Sequence)**.** Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. The number L is a **cluster value** of (x_n) if for every $\varepsilon > 0$ and every $N \in \mathbb{N}$ there exists $n \geq N$ such that

$$|x_n - L| < \varepsilon.$$

Theorem 0.39.2 (Cluster Values Are Subsequential Limits)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. Then L is a cluster value of (x_n) if and only if L is a subsequential limit of (x_n).*

Go to proof.

Theorem 0.39.3 (Bounded Sequences Have Cluster Values)**.** *Every bounded real sequence has at least one cluster value.*

Go to proof.

Theorem 0.39.4 (Limsup and Liminf Are Extremal Cluster Values)**.** *Let (x_n) be a bounded real sequence. Then $\limsup_{n\to\infty} x_n$ is the largest cluster value of (x_n), and $\liminf_{n\to\infty} x_n$ is the smallest cluster value of (x_n).*

Go to proof.

0.40 Cauchy

Cauchy Sequences

Definition 0.40.1 (Cauchy Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is a **Cauchy sequence** if

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \; \forall m, n \geq N \; \big(|x_m - x_n| < \varepsilon\big).$$

Theorem 0.40.2 (Convergent Sequences Are Cauchy)**.** *Let (x_n) be a real sequence. If (x_n) converges, then (x_n) is Cauchy.*

Go to proof.

Theorem 0.40.3 (Cauchy Sequences Are Bounded)**.** *Every Cauchy real sequence is bounded.*

Go to proof.

Theorem 0.40.4 (Cauchy Sequence with a Convergent Subsequence)**.** *Let (x_n) be a Cauchy real sequence. If a subsequence of (x_n) converges to $L \in \mathbb{R}$, then (x_n) converges to L.*

Go to proof.

Theorem 0.40.5 (Cauchy Criterion for Real Sequences). *Let (x_n) be a real sequence. Then (x_n) converges if and only if (x_n) is Cauchy.*

Go to proof.

Theorem 0.40.6 (Cauchy Criterion via Tails). *Let (x_n) be a real sequence. Then (x_n) is Cauchy if and only if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that the N-tail has pairwise term distances less than ε.*

Go to proof.

Theorem 0.40.7 (Cauchy Tail Diameter Criterion). *Let (x_n) be a real sequence. Then (x_n) is Cauchy if and only if every sufficiently late tail has arbitrarily small pairwise diameter.*

Go to proof.

Theorem 0.40.8 (Cauchy Sequences Have Vanishing Successive Differences). *If (x_n) is a Cauchy real sequence, then*

$$|x_{n+1} - x_n| \to 0.$$

Go to proof.

Theorem 0.40.9 (Scalar Multiples of Cauchy Sequences). *Let (x_n) be a Cauchy real sequence, and let $\alpha \in \mathbb{R}$. Then (αx_n) is Cauchy.*

Go to proof.

Theorem 0.40.10 (Sums of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences. Then $(x_n + y_n)$ is Cauchy.*

Go to proof.

Theorem 0.40.11 (Differences of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences. Then $(x_n - y_n)$ is Cauchy.*

Go to proof.

Theorem 0.40.12 (Linear Combinations of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences, and let $\alpha, \beta \in \mathbb{R}$. Then $(\alpha x_n + \beta y_n)$ is Cauchy.*

Go to proof.

Theorem 0.40.13 (Products of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences. Then $(x_n y_n)$ is Cauchy.*

Go to proof.

Theorem 0.40.14 (Reciprocals of Cauchy Sequences). *Let (x_n) be a Cauchy real sequence. If there exist $c > 0$ and $N_0 \in \mathbb{N}$ such that $|x_n| \geq c$ for every $n \geq N_0$, then $(1/x_n)$ is Cauchy.*

Go to proof.

Theorem 0.40.15 (Quotients of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences. If there exist $c > 0$ and $N_0 \in \mathbb{N}$ such that $|y_n| \geq c$ for every $n \geq N_0$, then (x_n/y_n) is Cauchy.*

Go to proof.

Theorem 0.40.16 (Absolute Values of Cauchy Sequences). *Let (x_n) be a Cauchy real sequence. Then $(|x_n|)$ is Cauchy.*

Go to proof.

0.41 Applications of Sequences

Applications of Sequences

Definition 0.41.1 (Newton Sequence for $\sqrt{2}$)**.** The **Newton sequence for** $\sqrt{2}$ is the real sequence (x_n) defined by

$$x_1 := \frac{3}{2}, \qquad x_{n+1} := \frac{1}{2}\left(x_n + \frac{2}{x_n}\right) \quad \text{for every } n \in \mathbb{N}.$$

Theorem 0.41.2 (Newton Approximation of $\sqrt{2}$)**.** *Let (x_n) be the Newton sequence for $\sqrt{2}$. Then (x_n) converges and*

$$\lim_{n\to\infty} x_n = \sqrt{2}.$$

Go to proof.

Definition 0.41.3 (Factorial Partial Sums)**.** The **factorial partial-sum sequence** is the real sequence (s_n) defined by

$$s_n := \sum_{k=0}^{n} \frac{1}{k!} \qquad \text{for every } n \in \mathbb{N}.$$

Theorem 0.41.4 (Factorial Partial Sums Approximate e)**.** *Let (s_n) be the factorial partial-sum sequence. Then (s_n) converges to the Euler number e:*

$$\lim_{n\to\infty} s_n = e.$$

Go to proof.

Definition 0.41.5 (Compound-Interest Sequence)**.** The **compound-interest sequence** is the real sequence (c_n) defined by

$$c_n := \left(1 + \frac{1}{n}\right)^n \qquad \text{for every } n \in \mathbb{N}.$$

Theorem 0.41.6 (Compound-Interest Approximation of e)**.** *Let (c_n) be the compound-interest sequence. Then*

$$\lim_{n\to\infty} c_n = e.$$

Go to proof.

Definition 0.41.7 (Decimal Truncation Sequence)**.** Let $\alpha \in \mathbb{R}$. The **decimal truncation sequence** of α is the real sequence (d_n) defined by

$$d_n := \frac{\lfloor 10^n \alpha \rfloor}{10^n} \qquad \text{for every } n \in \mathbb{N}.$$

Theorem 0.41.8 (Decimal Truncations Converge)**.** *Let $\alpha \in \mathbb{R}$, and let (d_n) be the decimal truncation sequence of α. Then*

$$\lim_{n\to\infty} d_n = \alpha.$$

Go to proof.

Proofs

Capstone

Series

0.42 Finite Series

Definition 0.42.1 (Finite Real Sequence on an Integer Interval)**.** Let $m, n \in \mathbb{Z}$ with $m \leq n$. A finite real sequence indexed from m to n is an assignment of a real number a_i to each integer i satisfying $m \leq i \leq n$. We denote such a sequence by $(a_i)_{i=m}^n$.

Definition 0.42.2 (Empty Finite Sum)**.** Let $m, n \in \mathbb{Z}$. If $n < m$, the finite sum from m to n is defined by

$$\sum_{i=m}^{n} a_i := 0.$$

Definition 0.42.3 (Finite Sum)**.** Let $m, n \in \mathbb{Z}$, and let real numbers a_i be assigned for the relevant integer indices. The notation

$$\sum_{i=m}^{n} a_i$$

denotes the finite sum, or finite series, of the terms indexed from m to n.

Definition 0.42.4 (Recursive Finite Sum)**.** Let $m \in \mathbb{Z}$, and let real numbers a_i be assigned for the relevant indices. The finite sum, also called a finite series, is defined recursively from the empty finite sum by the rule

$$\sum_{i=m}^{n+1} a_i := \left(\sum_{i=m}^{n} a_i \right) + a_{n+1} \qquad \text{whenever } n \geq m - 1.$$

Lemma 0.42.5 (Splitting a Finite Sum)**.** *Let $m \leq n < p$ be integers, and let a_i be a real number assigned to each integer i satisfying $m \leq i \leq p$. Then*

$$\sum_{i=m}^{n} a_i + \sum_{i=n+1}^{p} a_i = \sum_{i=m}^{p} a_i.$$

Go to proof.

Definition 0.42.6 (Summation over a Finite Set)**.** Let X be a finite set with n elements, where $n \in \mathbb{N}$, and let $f : X \to \mathbb{R}$. Choose a bijection

$$g : \{i \in \mathbb{N} : 1 \leq i \leq n\} \to X.$$

The finite sum of f over X is defined by

$$\sum_{x \in X} f(x) := \sum_{i=1}^{n} f(g(i)).$$

Proposition 0.42.7 (Finite Summations Are Well-Defined)**.** *Let X be a finite set with n elements, where $n \in \mathbb{N}$, let $f : X \to \mathbb{R}$, and let*

$$g : \{i \in \mathbb{N} : 1 \leq i \leq n\} \to X \qquad \text{and} \qquad h : \{i \in \mathbb{N} : 1 \leq i \leq n\} \to X$$

be bijections. Then

$$\sum_{i=1}^{n} f(g(i)) = \sum_{i=1}^{n} f(h(i)).$$

Go to proof.

Proposition 0.42.8 (Empty Finite Set Summation)**.** *If X is empty, and $f : X \to \mathbb{R}$ is a function, then*

$$\sum_{x \in X} f(x) = 0.$$

Go to proof.

Proposition 0.42.9 (Singleton Finite Set Summation)**.** *If X consists of a single element, $X = \{x_0\}$, and $f : X \to \mathbb{R}$ is a function, then*

$$\sum_{x \in X} f(x) = f(x_0).$$

Go to proof.

Proposition 0.42.10 (Substitution for Finite Set Summation)**.** *If X is a finite set, $f : X \to \mathbb{R}$ is a function, and $g : Y \to X$ is a bijection, then*

$$\sum_{x \in X} f(x) = \sum_{y \in Y} f(g(y)).$$

Go to proof.

Proposition 0.42.11 (Integer Interval as a Finite Set)**.** *Let $n \leq m$ be integers, and let*

$$X := \{i \in \mathbb{Z} : n \leq i \leq m\}.$$

If a_i is a real number assigned to each integer $i \in X$, then

$$\sum_{i=n}^{m} a_i = \sum_{i \in X} a_i.$$

Go to proof.

Proposition 0.42.12 (Disjoint Union Finite Set Summation)**.** *Let X and Y be disjoint finite sets, so $X \cap Y = \emptyset$, and let $f : X \cup Y \to \mathbb{R}$ be a function. Then*

$$\sum_{z \in X \cup Y} f(z) = \left(\sum_{x \in X} f(x) \right) + \left(\sum_{y \in Y} f(y) \right).$$

Go to proof.

Proposition 0.42.13 (Addition Rule for Finite Set Summation)**.** *Let X be a finite set, and let $f : X \to \mathbb{R}$ and $g : X \to \mathbb{R}$ be functions. Then*

$$\sum_{x \in X} (f(x) + g(x)) = \sum_{x \in X} f(x) + \sum_{x \in X} g(x).$$

Go to proof.

Proposition 0.42.14 (Scalar Multiplication Rule for Finite Set Summation)**.** *Let X be a finite set, let $f : X \to \mathbb{R}$ be a function, and let c be a real number. Then*

$$\sum_{x \in X} cf(x) = c \sum_{x \in X} f(x).$$

Go to proof.

Proposition 0.42.15 (Monotonicity for Finite Set Summation)**.** *Let X be a finite set, and let $f : X \to \mathbb{R}$ and $g : X \to \mathbb{R}$ be functions such that $f(x) \leq g(x)$ for all $x \in X$. Then*

$$\sum_{x \in X} f(x) \leq \sum_{x \in X} g(x).$$

Go to proof.

Proposition 0.42.16 (Triangle Inequality for Finite Set Summation)**.** *Let X be a finite set, and let $f : X \to \mathbb{R}$ be a function. Then*

$$\left| \sum_{x \in X} f(x) \right| \leq \sum_{x \in X} |f(x)|.$$

Go to proof.

Lemma 0.42.17 (Iterated Summation over a Product)**.** *Let X and Y be finite sets, and let $f : X \times Y \to \mathbb{R}$ be a function. Then*

$$\sum_{x \in X} \left(\sum_{y \in Y} f(x, y) \right) = \sum_{(x,y) \in X \times Y} f(x, y).$$

Go to proof.

Corollary 0.42.18 (Fubini's Theorem for Finite Series)**.** *Let X and Y be finite sets, and let $f : X \times Y \to \mathbb{R}$ be a function. Then*

$$\sum_{x \in X} \left(\sum_{y \in Y} f(x, y) \right) = \sum_{(x,y) \in X \times Y} f(x, y) = \sum_{(y,x) \in Y \times X} f(x, y) = \sum_{y \in Y} \left(\sum_{x \in X} f(x, y) \right).$$

Go to proof.

Lemma 0.42.19 (Reindexing a Finite Sum)**.** *Let $m \leq n$ be integers, let k be an integer, and let a_i be a real number assigned to each integer i satisfying $m \leq i \leq n$. Then*

$$\sum_{i=m}^{n} a_i = \sum_{j=m+k}^{n+k} a_{j-k}.$$

Go to proof.

Lemma 0.42.20 (Finite Sum of Sums)**.** *Let $m \leq n$ be integers, and let a_i and b_i be real numbers assigned to each integer i satisfying $m \leq i \leq n$. Then*

$$\sum_{i=m}^{n} (a_i + b_i) = \left(\sum_{i=m}^{n} a_i \right) + \left(\sum_{i=m}^{n} b_i \right).$$

Go to proof.

Lemma 0.42.21 (Finite Sum of Scalar Multiples)**.** *Let $m \leq n$ be integers, let a_i be a real number assigned to each integer i satisfying $m \leq i \leq n$, and let $c \in \mathbb{R}$. Then*

$$\sum_{i=m}^{n}(ca_i) = c\left(\sum_{i=m}^{n} a_i\right).$$

Go to proof.

Lemma 0.42.22 (Triangle Inequality for Finite Series)**.** *Let $m \leq n$ be integers, and let a_i be a real number assigned to each integer i satisfying $m \leq i \leq n$. Then*

$$\left|\sum_{i=m}^{n} a_i\right| \leq \sum_{i=m}^{n}|a_i|.$$

Go to proof.

Lemma 0.42.23 (Comparison Test for Finite Series)**.** *Let $m \leq n$ be integers, and let a_i and b_i be real numbers assigned to each integer i satisfying $m \leq i \leq n$. Suppose that $a_i \leq b_i$ for all i satisfying $m \leq i \leq n$. Then*

$$\sum_{i=m}^{n} a_i \leq \sum_{i=m}^{n} b_i.$$

Go to proof.

0.43 Infinite Series

Definition 0.43.1 (Formal Infinite Series)**.** A formal infinite series is an expression of the form

$$\sum_{n=m}^{\infty} a_n,$$

where m is an integer, and a_n is a real number for every integer $n \geq m$. This series is sometimes written as

$$a_m + a_{m+1} + a_{m+2} + \cdots.$$

Proofs

Function Sequences

Proofs